

the evolving trends within the current research domain, thereby enabling the LLM to better capture future development trends. The quality of generated ideas levels off after reaching a maximum length of 5. This saturation point indicates that this length is sufficient to capture relevant trends, with additional literature offering diminishing returns.

4.6 WIDTH OF CoI

We also assess the impact of the width of CoI (i.e., the branch number K) on the quality of generated ideas. Figure 7 shows the trend of average ELO scores with varying branch numbers. Generally, increasing the branch numbers shows a positive correlation with idea quality. However, the disparity in ELO scores across different branch numbers is small. This phenomenon can likely be attributed to the fact that generating multiple chains primarily helps reduce the impact of any single CoI performing poorly. Fortunately, such low-quality CoIs are rare.

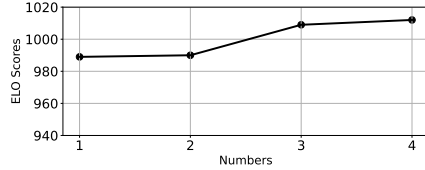


Figure 7: Width analysis of the CoI.

5 RELATED WORKS

Scientific Research Idea Generation. Idea generation is a fundamental step in scientific research. Due to its innovative nature, idea generation has been primarily a human-driven activity. However, recent studies indicate that LLMs can generate plausibly novel and feasible ideas as those of human researchers (Si et al., 2024; Kumar et al., 2024). To investigate the potential of LLMs in idea generation, prior works begin with the task of scientific hypothesis discovery (Yang et al., 2024b; Qi et al., 2023; Wang et al., 2023), which aims to elucidate relationships between two scientific variables. Despite its utility, scientific hypothesis discovery may not fully capture the complexity and multifaceted nature of real-world problems. To address this limitation, projects like GPT-Researcher (Assafelovic, 2023) and ResearchAgent (Baek et al., 2024) have adopted a more open-ended idea generation scenario including the underlying methodologies and experimental designs. They leverage agent-based systems to enhance the quality of idea generation. Beyond ideation, numerous studies also explore the use of LLMs for executing experiments (Huang et al., 2024; Tian et al., 2024) or combining both idea generation and experimental execution (Li et al., 2024; Lu et al., 2024). However, these approaches often make minor modifications to existing ideas for drafting their ideas, which often lack depth and creativity.

Align LLMs with Human Cognitive Patterns. As LLMs are trained with vast amounts of human data (Brown et al., 2020), this may enable them to internalize human cognitive patterns. Firstly, CoT (Wei et al., 2022) indicates that LLMs can enhance their reasoning abilities when provided with step-by-step guidance. Further research supports this notion by showing that simply prompting LLMs to engage in step-by-step reasoning can trigger better reasoning capability (Kojima et al., 2022). Additionally, Fu et al. (2022) reveals that in-depth reasoning of LLMs can be achieved with more elaborate prompts. As a result, a prompting strategy that closely emulates human cognition is likely to elicit more insightful responses from these models. Motivated by this, we propose CoI to better mimic the progressive cognitive patterns of humans when generating new research ideas.

6 CONCLUSIONS

In this paper, we introduce Chain of Ideas (CoI) agent, a framework designed to enhance the capability of LLMs in generating research ideas. The CoI agent offers a promising and concise solution by organizing ideas into a chain structure, effectively mirroring the progressive development within a given research domain. It facilitates LLMs to digest the current advancements in research, thereby enhancing their ideation capabilities. To comprehensively evaluate the capability of automated idea generation methods, we also propose Idea Arena, an evaluation system that requires the participant methods to compete in pairs about their generated ideas for the research topics, which demonstrates high agreement with human evaluation. Experimental results indicate that the CoI agent consistently outperforms other methods and is capable of generating ideas comparable to human creativity.

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